



AVINYA

2026

By Prakriti Club

Prakriti × Techniche
**Ecolnnovate
Challenge**

Where Ideas Meet Impact.

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Welcome to **AVINYA 2026**

Get ready to embark on a journey of innovation with AVINYA 2026, the flagship eco-hackathon by the Prakriti Club of IIT Guwahati. This platform is designed to empower creative minds to tackle real-world environmental challenges through innovation and technology. With prizes worth ₹1,00,000 up for grabs, AVINYA invites young thinkers to showcase their talent and make a tangible impact on the planet.



About Us

Prakriti Club is the techno-environmental wing of IIT Guwahati, dedicated to crafting sustainable solutions for environmental issues. From devising carbon emission calculators for the campus to building smart polyhouse models, crop health detection, our club has continually explored the intersection of technology and ecology.

We also run awareness campaigns, plantation drives, and community events that echo our commitment to sustainability. Partnering with Techniche, IIT Guwahati's renowned techno-management festival, Prakriti aims to inspire large-scale participation and cross-disciplinary innovation. Techniche has touched the lives of over a million people through competitions, social initiatives, and its marathon events. Together, AVINYA 2026 is our initiative to unite innovators, engineers, and changemakers to build practical solutions for a greener future.



Avinya 2026 will unfold in two dynamic phases:

Phase 1: Participants will be provided with a curated list of nine impact problem statements across various sustainability and innovation domains. Teams must choose one problem statement, brainstorm and design a practical solution, and submit their ideas in a structured presentation format by the deadline.

Phase 2: Shortlisted teams will be invited to the IIT Guwahati campus to showcase their ideas during the institute's flagship festival. They will then engage in a 24–30 hour on-site innovation sprint, tackling new problem statements introduced by industry mentors and domain experts.

Interactive Sessions & Expert Talks: Throughout the event, participants will benefit from engaging workshops and insightful guest lectures conducted by professionals from academia and industry to deepen understanding and enhance the quality of their solutions.



Software & Web Development

1. Peer-to-Peer Microgrid Dashboard

- **Explanation:** As the adoption of small-scale distributed energy resources like rooftop solar and localized biogas grows, communities have the opportunity to trade surplus energy locally. However, without accessible digital platforms to track generation, consumption, and trading, managing these microgrids remains difficult for the average energy prosumer.
- **Objective:** Develop a responsive web application that visualizes real-time energy generation and consumption for a community microgrid, enabling neighbors to conceptually trade surplus energy.
- **Goal:** To build an intuitive digital marketplace that empowers communities to manage their distributed energy resources effectively, incentivizing localized renewable energy adoption and grid independence.



2. Urban Carbon Footprint Calculator

- **Explanation:** Universities and urban communities generate significant greenhouse gas emissions through daily activities such as transportation, energy use, and waste disposal. While quantifying these emissions is a crucial step toward sustainability, individuals often lack accessible, localized tools to measure and understand their personal environmental impact.
- **Objective:** Build a lightweight, interactive digital tool tailored for university campuses that allows individuals to input their daily habits to visualize their carbon footprint and receive automated reduction strategies.
- **Goal:** To empower students and faculty with clear, data-driven insights about their personal environmental impact, fostering a culture of sustainability and encouraging actionable behavioral changes across the campus.



3. Smart Polyhouse Management Interface

- **Explanation:** Urban agriculture often relies on polyhouses to maintain controlled environments for optimal crop growth. However, manual monitoring of internal conditions is labor-intensive and prone to human error, often leading to wasted water or poor crop health. Developing smart polyhouse models is a proven way to explore the intersection of technology and ecology.
- **Objective:** Create a digital solution that processes simulated temperature, soil moisture, and humidity data to automatically recommend ventilation and watering schedules for urban polyhouse farming.
- **Goal:** To increase agricultural efficiency and crop yield by providing an accessible, digital management tool that minimizes water waste and manual labor in urban farming environments.



Computational Modelling & Simulation

1. Stationary Solar Concentrator Design

- **Explanation:** Decentralized solar thermal applications, such as residential water heating, often rely on complex moving parts for active sun-tracking to achieve necessary temperatures. These mechanical tracking systems are expensive, bulky, and prone to maintenance failures, making them less accessible for widespread, reliable community deployment.
- **Objective:** Design and thermally simulate a small-scale, stationary solar concentrator to provide sustainable and cost-effective thermal energy for residential water heating.
- **Goal:** To engineer a reliable, static thermal harvesting system that reduces reliance on energy-intensive heating while eliminating the structural and financial burdens of mechanical tracking components.



2. Urban Heat Island Mitigation Simulation

- **Explanation:** Urban heat islands worsen with increasing temperatures and reduced green cover. Smart cooling strategies are essential for reducing heat stress, energy consumption, and improving quality of life. Modelling these environments is a critical step to determine the most effective interventions before physical construction begins.
- **Objective:** Create a computational simulation to evaluate and optimize the strategic placement of localized water bodies, green infrastructure, and high-albedo building materials to reduce ambient temperatures in a densely populated city block.
- **Goal:** To develop a data-driven, localized mitigation strategy that city planners can use to build healthier, cooler urban environments and reduce dependence on energy intensive cooling systems.



Rapid Design & Systems Engineering

1. Hydrogen Fuel Retrofit Roadmap

- **Explanation:** Small-scale internal combustion engines are significant contributors to localized carbon emissions. Transitioning existing engine hardware to utilize zero emission fuels offers a faster, more cost-effective bridge to clean transportation and power generation than completely replacing systems with entirely new electric infrastructure.
- **Objective:** Develop a comprehensive physical retrofit framework and safety protocol that enables standard, small-scale internal combustion engines to safely and efficiently utilize hydrogen gas as their primary fuel source.
- **Goal:** To provide a safe, actionable, and scalable engineering roadmap for repurposing existing fossil-fuel engines into clean, hydrogen-powered systems without requiring complete hardware replacement.



2. Mechanical Biogas Flow and Purification Manifolds

- **Explanation:** Local organic waste digesters have enormous potential for community energy generation. However, processing and distributing this gas safely typically requires expensive, fragile electronic sensors for pressure regulation and purification, which can be costly and are prone to failure in raw environmental conditions.
- **Objective:** Engineer a strictly mechanical pressure regulation and scrubbing unit that connects local organic waste digesters to a communal energy-sharing network.
- **Goal:** To create a robust, low-maintenance hardware solution that enables safe, efficient, and decentralized waste-to-energy conversion for community distribution.



3. Smart Renewable Energy Solutions Without Electronics

- **Explanation:** Many energy systems rely heavily on electronics, which are costly, hard to maintain, and contribute to e-waste. Mechanical or material-based smart solutions can be a sustainable alternative.
- **Objective:** Develop innovative renewable energy systems that operate intelligently without electronic components.
- **Approach:** Utilize material science or mechanical principles to create energy systems that self-adjust or optimize energy generation and storage without digital controllers.



4. Decentralized Smart Waste-to-Energy Collection Network

- **Explanation:** Standard municipal waste collection is highly centralized, carbon intensive, and often fails to properly segregate organic waste at the source. This misses a massive opportunity for local energy generation. A smart, localized collection approach can turn everyday community waste into an active energy asset.
- **Objective:** Develop a smart, community-level waste sorting and collection framework that seamlessly tracks and routes organic waste directly to local biogas digesters for immediate energy conversion.
- **Goal:** To incentivize proper waste segregation at the source, reduce the logistical burden on municipal landfills, and actively power community-based renewable energy initiatives.



Timeline & Event Structure

Timeline

- **Launch of Problem Statements**
Date - 31th July 2026
- **Deadline for Submission**
Date -15th August 2026 (11:59 PM IST)
- **Announcement of Shortlisted Teams**
Date - : 20th August 2026
- **Final On-Campus Event**
Date - : 27th - 30th August 2026



Round 1

- Participants will receive eight carefully curated problem statements addressing sustainability, technology, and social innovation
- Teams must select one problem statement to work on
- They will develop a solution proposal and presentation (PPT format) based on their research, creativity, and feasibility
- Submissions will be evaluated by a panel of experts on parameters such as innovation, impact, scalability, and design clarity



Round 2

- Top-performing teams from Round 1 will be invited to IIT Guwahati to participate in an intensive on ground challenge
- Teams will present their initial solutions live to an expert jury
- A 24–30 hour hackathon-style sprint will follow, where teams tackle new real-world challenges shared by industry partners
- The final round emphasizes collaboration, rapid prototyping, and presentation skills under mentorship and time pressure.



Marking Schemes

Round 1:

- Solution Innovation: 30%
- Technical Feasibility: 25%
- Scalability and Practicality: 25%
- Sustainability Impact: 20%

Round 2:

- Phase-1: 50%
- Phase-2: 50%



Rewards & Incentives

Prize Pool:

₹1,00,000 (including ₹65,000 cash)

Exciting Goodies:

Branded items and useful tools

Certificates:

For participation and winning teams

Networking Opportunities:

Engage with industry leaders and experts

Workshops & Sessions:

Learn directly from specialists in the field



General Rules

- Each team must consist of 2 to 5 members
- Submissions must be original and adhere to deadlines
- Plagiarism or dishonesty will lead to disqualification
- Judging decisions made by the Prakriti team and experts are final
- Submission deadline for Round 1: 15th August 2026



Who Can Participate?

Students currently enrolled in Bachelor's, Master's, or PhD programs from any academic institution are welcome to participate.

Whether you're from engineering, science, design, or management—if you have a passion for sustainability and innovation, AVINYA is for you.



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